WASTEWISE SCAN · DESIGN PRECIS

WasteWise Scan design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: fixes F1 to F3 of WSC-DEC-001 (baffle through the glass with a separate detector window, four-cycle scan, 11 x 10 mm cell strap); numbers from WSC-CAL-001 v0.5 (0.64 s, 21.9 h, 239 g and 259 g with cap); requirements now PR numbers (WSC-PRS-001); concept options WSC-OPT-001 and STEM check WSC-STM-001
0.7	2026-10-02	Amish Chadha	Mass restated for 2.0 mm shell walls (238 g, 258 g with cap); paper band study result; numbers from WSC-CAL-001 v0.4
0.6	2026-10-02	Amish Chadha	Open questions answered by Amish's decisions of 2026-10-02 (WSC-DEC-001)
0.5	2026-10-02	Amish Chadha	Constructable design (WSC-DDR-004) and build plan WSC-BLD-001; components, mass and cost from WSC-CAL-001 v0.3; budget as a value-engineering target
0.4	2026-09-25	Amish Chadha	

			Recommendations accepted by Amish (DDR-002); WSC-DDR-003 applied (black cap as backing, dark-level prompt, sun hood, open-air crosstalk reading, shared record); numbers from WSC-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3; decisions of WSC-DDR-001 applied; numbers from WSC-CAL-001; 47 kOhm gain and interleaved dark readings; new findings on sunlight, display and window crosstalk
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise Scan design precis

WasteWise Scan is a palm-sized scanner that the user presses against a plastic item. It flashes eight near-infrared bands one after another, reads the reflection with an InGaAs photodiode, and in about 0.64 s shows the resin, a confidence mark and the local price grade. The calculation note WSC-CAL-001 v0.5 shows that it meets the speed, battery, mass, sunlight, logging, calibration, eye-safety and prototype cost requirements on paper. Clear and translucent items are backed with the black calibration cap in sun, and a clip-on sun hood shades the display (decisions accepted by Amish on 2026-09-25, WSC-DDR-003). Direct-sun legibility (R7) is at risk because the hood cannot shade every sun angle, and resin accuracy (R1) remains at risk near 1,700 nm. Value-engineering target: USD 150. Estimated cost of the constructable design: USD 167.50 (USD 17.50 over the target). Since 2026-10-02 the design is constructable (WSC-DDR-004, open for Amish's review): screws, a gasket, an optical head block, a cell cradle and strap, a display frame and clip legs on the hood were added, and the scanner now weighs about 239 g (259 g with the cap) with the 2.0 mm shell walls Amish decided on 2026-10-02, so the 250 g mass requirement (PR-006, was R6) is not met, 9 g over with the cap. On 2026-10-24 the STEM check (WSC-STM-001) and its independent review led to three fixes made under Amish's standing instruction of 2026-10-23 (WSC-DEC-001 F1 to F3): the head block's baffle tube now passes through the glass and holds a separate detector window, so LED light reflected inside the window cannot reach the detector; the scan repeats its dark-band-dark sequence four times; and the cell strap is larger. The current design is option A of the concept options WSC-OPT-001; option B, a tunable-filter spectrometer, scores within 10 % of it, so the choice is with Amish. Requirements are now numbered PR-001 to PR-017 (WSC-PRS-001; R1 to R14 map to PR-001 to PR-014). The prototype build plan is WSC-BLD-001 (05-build-plan.md).

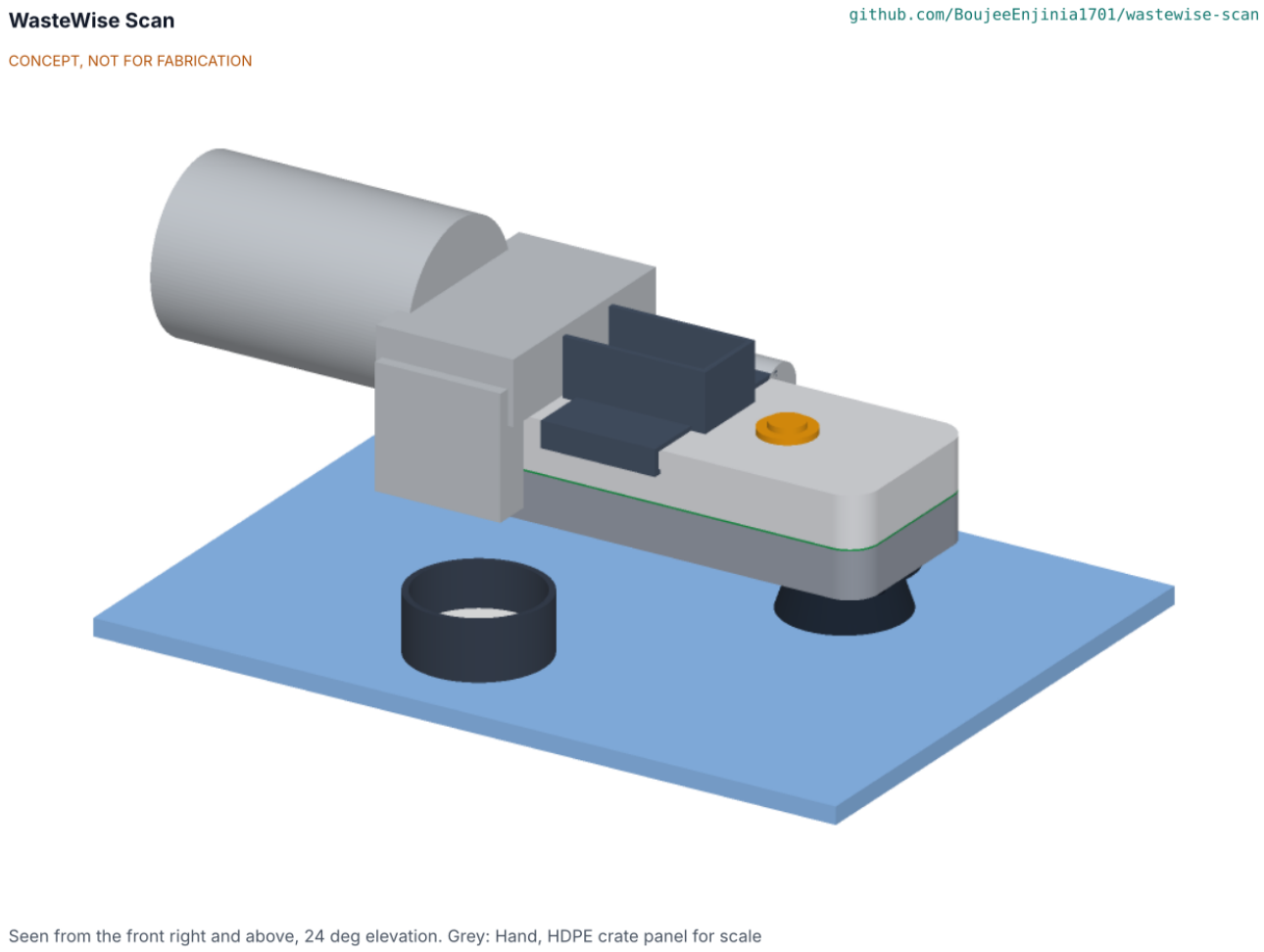


Figure 1. Concept render from the parametric model. The grey hand and the blue crate panel are for scale only.

The sensor

The scaffold named an 18-channel AS7265x-class sensor. That chip covers about 410 to 940 nm (SparkFun Triad board), which misses the polymer absorption bands near 1,150 to 1,250 nm and 1,650 to 1,750 nm that NIR sorters rely on (see WSC-PRB-001 and the review by Neo et al. 2022). Amish decided on 2026-09-25 (WSC-DDR-001, D1) to use the approach of the open Plastic Scanner project instead: discrete NIR and short-wave infrared (SWIR) LEDs from 850 to 1,650 nm and one InGaAs photodiode (Hackaday, 2021). It is still a low-cost multispectral sensor, so the pitch is unchanged.

How it works

1. **Press.** The user presses the soft black shroud against a flat or gently curved part of the item. The shroud blocks sunlight falling on the spot and fixes the item 22 mm from the detector.
2. **Illuminate and read.** On a button press the firmware pulses each of eight LEDs in turn (850, 940, 1,050, 1,200, 1,300, 1,450, 1,550 and 1,650 nm; WSC-DDR-001 D4). Each band reading sits between two dark readings, so slow changes in ambient light are subtracted, and the whole sequence runs four times and is averaged (fix F2). If the mean dark level exceeds 1.24 times the weakest band reading, which happens when sunlight passes through an unbacked clear item, the scanner shows "shade or back with cap" instead of a result (WSC-DDR-003

D8). The user then holds the black calibration cap behind the item and scans again. A transimpedance amplifier with 47 kΩ feedback and a 24-bit ADC take two conversions per reading. Eight bands and nine darks take about 137 ms, and the four cycles about 548 ms.

3. **Normalize.** Readings have the remaining crosstalk offset from the last open-air reading subtracted (WSC-DDR-003 D11; with fix F1 the baffle tube through the glass keeps reflections inside the window from reaching the detector) and are divided by the last white reference taken in the calibration cap, giving a reflectance per band that is independent of LED aging and, to first order, temperature. A board temperature reading flags drift of more than 5 %, which takes about 12.5 °C of change.
4. **Classify.** A small classifier on the ESP32-S3 (linear discriminant analysis or a small decision tree on normalized band ratios; WSC-DDR-001 D5) returns a resin and a confidence. Below the confidence threshold, or when mean reflectance is under about 0.08 (black items), it shows "unknown".
5. **Show and log.** The display shows resin code, name, color and icon, plus the grade from the local price table. Each scan is logged as a 64-byte record for export over USB as CSV or JSON Lines.
6. **Hand off.** Unknown or low-confidence items can be photographed with a phone and passed to WasteWise-ml. The shared record proposed in WSC-DDR-002 lets the two results be joined on a pair ID.

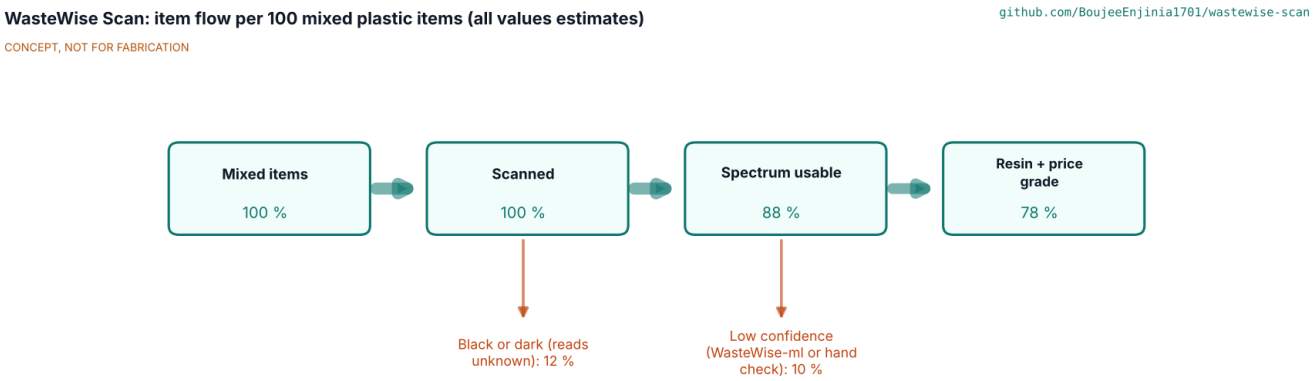


Figure 2. Item flow per 100 mixed plastic items. All values are estimates for concept review.

Main components

Table 1. Main components. Numbers match the BOM, the exploded view and drawing WSC-DWG-001.

#	Component	Choice	Notes
1	Top shell	3D-printed PETG with display window, button hole and eight insert bosses	Four M3 screws from below; foam gasket on the rim; about 51 g
2	Controller and display	ESP32-S3 board with 1.9 in IPS display and on-board Li-ion charger (LilyGO T-Display-S3 class)	No custom PCB (WSC-DDR-001 D3)
3	LED ring	Eight LEDs on a 20 mm pitch circle in a printed optical head block, each tilted about 23° so the beams meet on the item	SWIR LEDs dominate cost; the block also seats the photodiode and carries item 5
4	Photodiode	InGaAs PIN photodiode, 1 mm, TO-46, about 900 to 1,700 nm, inside a short baffle tube	Standard InGaAs, cutoff near 1,700 nm
5	Amplifier and ADC	Low-noise op-amp transimpedance stage with 47 kΩ feedback, 24-bit ADC (ADS1220 class), LED switches	On perfboard for the first build
6	Windows	Borosilicate LED window, 25 mm, 2 mm, with an 8.4 mm cored centre hole; separate 5.8 mm detector window in the end of the head block's baffle tube (fix F1)	Keep dust out; uncoated; the tube through the glass blocks crosstalk
7	Light shroud	Black TPU cone, 18 mm deep, 42 mm across the rim, 36 mm inside, with a 47 mm clamping flange	Blocks direct light, sets distance; its three screws clamp the window and head block
8	Battery	One protected 18650 Li-ion cell, 3,000 mAh	WSC-DDR-001 D7
9	Cell contacts and USB-C	Spring and flat contacts in a cradle printed with the bottom shell; USB-C power board in the tail wall	A printed strap (item 18) holds the cell against about 550 N on a corner drop
10	Bottom shell and grip	3D-printed PETG, textured grip, window hole, cell cradle, screw bosses	About 56 g
11	Scan button	Large sealed 16 mm push button for gloved hands	
12, 13	Calibration cap	Printed black PETG cap, 48 mm OD, with a 38 mm white PTFE reference disc; stores over the shroud	White reference (WSC-DDR-001 D6); also the backing for clear items in sun (WSC-DDR-003 D9)
14	Sun hood	Clip-on printed PETG hood, walls 20 mm high on both sides and at the head end of the display window, open toward the user; legs snap under ribs on the top shell	About 11 g and $1 (WSC-DDR-003 D10, WSC-DDR-004)
17	Display frame	Printed PETG frame holding the display board under the window	Four M3 screws into the top shell
18	Cell strap	Printed PETG bridge 11 x 10 mm across the cell (fix F3)	Two M3 screws in counterbores into floor bosses [J4]
19	Shell gasket	Closed-cell foam tape on the bottom shell rim	Squeezed to 0.6 mm; the bosses set the squeeze

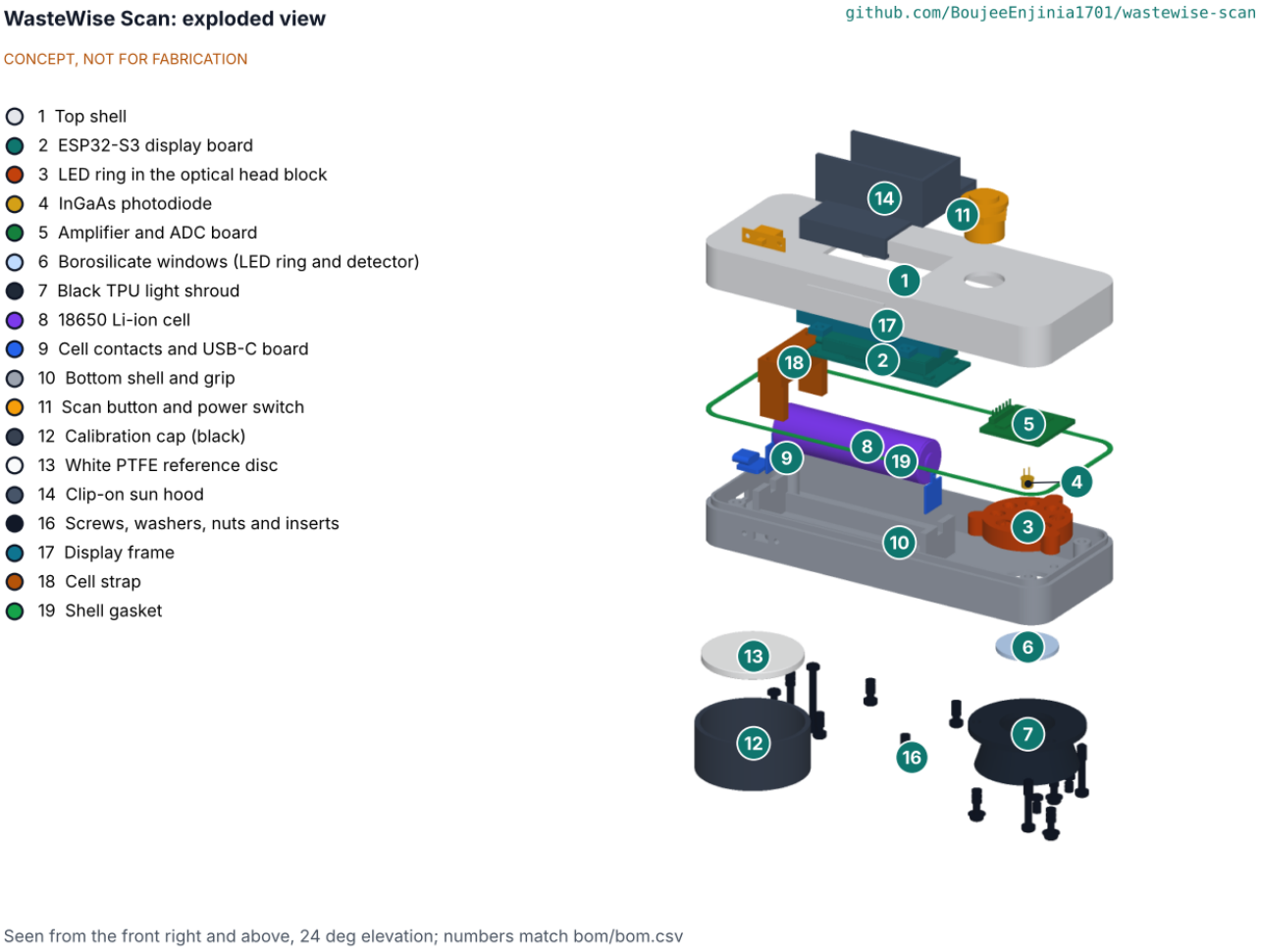


Figure 3. Exploded view with BOM numbers.

Key numbers

All values are from WSC-CAL-001 (tags in brackets) and rest on the assumptions stated there.

Table 2. Key numbers.

Quantity	Value	Requirement
Scan time	about 0.64 s [E2]	PR-002 (1.5 s) met
Detected signal on the white reference	0.35 to 5.2 µW, 10 to 115 mV at 47 kΩ [C2]	
Worst signal-to-noise ratio in shade	about 640 (clear item, 1,650 nm, four cycles) [C4]	600 needed (3 sigma at 0.5 %)
Sunlight through a clear item	about 23 µA, about 900 times the weakest signal [D1, D3]	Refused by the dark-level prompt [D7]
Ambient error, clear item backed with the cap	0.21 % of signal [D6]	PR-004 met by estimate
Average power	about 0.40 W [F2]	
Battery life	about 21.9 h [F2]	PR-005 (8 h) met
Log capacity	about 131,000 scans at 64 B [G1]	R9 met
Mass	about 239 g, 259 g with cap [A3]	**PR-006 (250 g) not met**, 9 g over
Size	160 x 62 x 34 mm (6.3 x 2.4 x 1.3 in), 72 mm high with shroud and sun hood, 68 mm wide over the hood legs [A1]	R6 size met
Display contrast in 100 klx sun	about 1.6:1 unshaded; 5.2:1 where the hood shades the screen [I1, I3, I4]	**R7 at risk**
LED irradiance at 200 mm	3.9 W/m² against 100 W/m² [H2]	R14 met by estimate
Parts cost	USD 167.50 [K2]	USD 17.50 over the USD 150 value-engineering target

Key design choices

- **Discrete LEDs and one InGaAs photodiode rather than a spectrometer or a visible-range chip** (WSC-DDR-001 D1). A SWIR spectrometer module costs far more than the budget; a visible-range chip misses the polymer bands. Eight bands are kept: the paper band study (WSC-CAL-001 v0.4, B4) scores a six-band set at 88 % against 89.5 % for eight on approximate reference spectra, which points to the six-band saving, but both are short of 95 % at the assumed item scatter, so the choice waits for measured items.
- **Contact measurement through a shroud.** Pressing on the item fixes the geometry and removes direct light on the spot. It does not stop light passing through clear items, so clear and translucent items in sun are backed with the black cap and the firmware refuses a result when the dark level shows through-light (WSC-DDR-003 D8, D9).
- **Low gain and interleaved darks.** The 47 kΩ gain keeps sunlight through clear items inside the ADC range; interleaved dark readings cancel slow ambient changes. Faster changes are handled by the backing step; LED modulation at 2 kHz was checked on paper and does not remove the need for it (WSC-CAL-001 D8).
- **Say "unknown" rather than guess.** A wrong call can contaminate a bale; the confidence threshold is set for few false results, accepting more unknowns (R3).

- **White reference in the protective cap** (D6). The user calibrates by pressing the scanner into its own cap, at the start of a shift and again when the drift warning shows; the same routine takes an open-air reading to remove window crosstalk. The cap is printed in black so it can also back clear items.
- **Local price table on the device.** Grades are what matter to users, and prices are local and change often. The cooperative edits them; no cloud service is needed.
- **Clip-on sun hood** (WSC-DDR-003 D10). A printed hood shades the display for most sun angles at negligible cost; large color and icon cues stay the primary result.
- **One common 18650 cell** (D7). Available and replaceable in most markets; the board's charger keeps the build simple.



Figure 4. Cutaway: LED ring, photodiode, amplifier board and window above the shroud, with the cell and controller behind.

Findings from the calculations and decisions taken

- **Clear items in sun (R4, now met by estimate).** Light passing through a clear or translucent item reaches the detector from inside the shroud, and hand movement changes it faster than dark subtraction can follow. Decided (WSC-DDR-003 D8, D9): back such items with the black cap (0.21 % error with the four-cycle scan of fix F2) and refuse a result when the dark level shows through-light. The classifier needs training data in this backed mode.

- **Display in direct sun (R7, now at risk).** A 400 cd/m² IPS screen washes out in 100 klx sun. Decided (D10): a clip-on sun hood, which gives about 5.2:1 on the shaded part but cannot shade the screen with the sun near its normal or over the open side.
- **Window crosstalk.** LED light reflected by the window's outer surface reaches the detector at about the same level as the item signal. Decided (D11): an open-air reading in the calibration routine subtracts it. Because a 10 % change from window dirt would leave about 10 % error, the STEM check failed this detail, and the fallback of a baffle extended through the glass (option (b)) is now in the design (fix F1, 2026-10-24): an 8 mm carbon-black tube passes through a cored hole in the LED window and holds a separate 5.8 mm detector window, with the gap sealed. The light left leaks through the tube wall, about 0.05 % of the signal on assumed values, and window dirt does not change it (WSC-CAL-001 C8); a printed coupon and the built head confirm it at TRL 4.
- **PE versus PP (R1 at risk).** The 1,650 nm band is clipped by the detector cutoff and only samples the short edge of the 1,650 to 1,750 nm region. Decided by Amish, 2026-10-02: keep eight bands, and add an extended InGaAs band only if the paper band study shows PE and PP cannot be told apart to R1 with eight (WSC-DEC-001). The study is done: PE against PP is 93 % with eight bands at the assumed item scatter, above 98 % at 1.0 % scatter, so it neither shows nor rules out the problem and the extended band is not added (WSC-CAL-001 v0.4, B5 to B7).

Safety

Safety: The scanner identifies plastic resin only. It must never be used to judge whether an item is safe to handle, or to classify hazardous, medical, chemical or pesticide containers. Waste handling carries cut, needle-stick and chemical risks: wear gloves and follow local guidance.

Safety: Backing a clear item with the calibration cap puts the user's hand close to the item. Do not back sharp, broken or contaminated items by hand; scan them in shade instead.

Safety: The 18650 lithium-ion cell stores about 11 Wh. Use a protected cell from a known supplier, charge with the board's charger only, do not charge in direct sun or above 45 °C, and stop using a cell that is dented, swollen or wet. The cradle must hold the cell firmly in a drop. Do not leave the device on hot surfaces or in closed vehicles.

Safety: The SWIR LEDs are invisible to the eye, so a user cannot tell when they are on. In normal use they fire for about 8 ms each, four times per scan, which keeps emission far inside the IEC 62471 exempt group by estimate (WSC-CAL-001, H). An LED stuck on at the shroud rim could exceed the corneal limit over long exposures, so the LED drivers need a hardware limit on on-time that firmware cannot override. Do not look into the window during a scan.

- Black and dark items read "unknown". Users must not treat "unknown" as any particular resin.
- The result is advice for sorting, not a certificate of material grade for sale.

Open questions

1. **First users and region** for co-design and the first price table. Decided by Amish, 2026-10-02: a member-owned waste picker cooperative that already sells sorted plastics to scrap buyers; the first candidate to approach is SWaCH in Pune, India, the partner chosen for WasteWise-ml, with Pune scrap prices as the first price table (WSC-DEC-001).

2. **Who the $150 volume target is for:** decided by Amish, 2026-10-02: cooperatives, buying a shared scanner (WSC-DEC-001).
3. **Extended InGaAs band.** Decided by Amish, 2026-10-02: eight bands; the band study comes first (WSC-DEC-001). The paper band study is done and is inconclusive; the extended band is not added and is raised as a proposal.
4. **Scan record format.** One shared record with a `pair_id` join was decided on this side on 2026-09-25 (WSC-DDR-002 v0.2, WSC-DDR-003 D12); WasteWise-ml has still to adopt it, and the pair ID is shown as a QR code on the scanner screen for the phone to scan, with digits as a fallback (decided by Amish, 2026-10-02, WSC-DEC-001).
5. **Band study.** A paper study on published reference spectra to confirm the band set and whether six bands suffice (WSC-DDR-001 D2, D4). Not yet done.

Concept media: blueprint sheet, interactive 3D model. General arrangement: WSC-DWG-001.